HydraDG: Governed Context Interventions with Fractal Custody for Agent Experiments

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Abstract

Agent evaluations increasingly depend on context interventions—retrieval, memory graphs, and prompt contracts—yet published reports often omit null outcomes, parser failures, and negative cells. We present HydraDG, a governed experimental framework that binds substantive model invocations, tool calls, and handoffs to *Fractal Custody Objects* (FCOs) and a *Fractal Custody Graph* (FCG). HydraDG preregisters hypotheses, freezes case manifests and scorers, records raw prompt and response hashes, and preserves terminal states including timeouts, abstentions, and malformed outputs without silent substitution. We report two preregistered studies (EXP-008, EXP-009) comparing flat prose versus structured FCG retrieval for failure-prevention endpoints; both terminate as *underpowered* under bounded case replication, so confirmatory effect claims are not supported while negative and null evidence remains in custody. We argue that custody-first evaluation infrastructure supports reproducible agent science rather than optional logging.

1 Introduction

Large language model agents are evaluated on task success, benchmark scores, and human preference [2, 3]. Less often preserved are the conditions under which experiments fail: swapped models, stale runtime selection, partial matrix execution, or JSON outputs that never parse. When negative and null cells disappear from the public record, later work cannot distinguish *no effect* from *no measurement*.

HydraDG addresses this gap with a custody-first experimental contract. Every actor—human operator, frontier agent, local model runtime, or deterministic script—must emit or be represented by a handoff receipt before its output is promoted. Receipts bind actor identity, host, repository state, input hashes, raw output hashes, evidence class, claim ceiling, and signature state (hashes are identity commitments, not digital signatures unless an authorized signing operation occurred). Probabilistic model outputs are never treated as authority; they are evidence objects subject to deterministic scoring and graph append rules.

This paper describes the HydraDG framework and reports terminal results from the IC Failure Learning structured-retrieval falsification lane. We do *not* claim that structured FCG context improves failure prevention; preregistered EXP-008 and EXP-009 closed as underpowered. We also do not treat exploratory secondary patterns as promoted findings.

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2 Related Work

Agent evaluation. Benchmark suites measure capability but rarely require publication of failed runs, malformed generations, or infrastructure blocks [2, 3]. HydraDG complements benchmarks with mandatory negative capture at the custody layer.

Retrieval and structured context. Retrieval-augmented generation and graph-augmented memory systems assume structured context helps [1, 4]. Falsification requires frozen contracts, explicit null retention, and power-aware verdicts; HydraDG enforces both.

Provenance and reproducibility. Preregistration, FAIR data principles, and nanopublication-style lineage support scientific reproducibility [5–7]. Agent stacks often stop at prompt logging. FCO/FCG extend content-addressed lineage to cross-actor handoffs so that a stale agent cannot write after ownership moves. Companion custody-framework preprints (deferred to camera-ready de-anonymization) define FCO/FCG formally; they establish framework provenance, not the HydraDG empirical results reported here.

3 Framework

3.1 Custody objects

An **FCO** (*Fractal Custody Object*) materializes one bounded transformation: e.g., a model call, scorer pass, or ingest operation. An **FCG** (*Fractal Custody Graph*) append records directed edges between FCOs with hash-linked roots. HydraDG preserves failures, nulls, and malformed outputs in custody (a failure-complete *behavior*, not a redefinition of FCO/FCG). Claim ceilings (exploratory mechanistic falsification, descriptive only, etc.) are declared at preregistration and enforced at promotion time.

3.2 Experimental freeze

Each experiment declares:

- frozen case manifest and prompt-contract hashes;
- conditions (e.g., flat prose vs. structured FCG);
- models with expected runtime digests;
- primary endpoint and aggregation rule;
- explicit gates for confirmatory vs. exploratory claims.

Execution emits `START_RECEIPT`, per-cell raw outputs with prompt/response SHA-256, parser state, and terminal receipts. Integrity failure (duplicate keys, missing dependencies, silent model substitution) blocks promotion; scientific failure (timeout, abstention, malformed JSON) is retained as terminal evidence.

3.3 Governed local execution

Where preregistered, local model calls route through a governed bridge that records selection age, swap posture, and resolved digest. Direct runtime invocation and governed invocation are separate evidence classes; one is not silently labeled as the other.

4 Experimental Setup

4.1 Task family

The IC Failure Learning suite uses synthetic agent-failure vignettes with preregistered endpoints such as E06 (prevents exposure of out-of-vault media). Cases are drawn from a frozen `CASES.jsonl` manifest; each case receives multiple replicates per condition.

Table 1: Terminal preregistered studies (HydraDG IC Failure Learning). *Underpowered* denotes insufficient paired evidence for confirmatory promotion, not proof of null effect.

Study	Primary verdict	Raw cells	Valid parse rate
EXP-008	UNDERPOWERED	300	0.907
EXP-009	UNDERPOWERED	300	0.883

4.2 Conditions and models

C0 (FLAT_PROSE): failure-handling guidance as unstructured text.
C1 (STRUCTURED_FCG): the same semantic content composed through a frozen FCG retrieval contract.

EXP-008 and EXP-009 used local models `qwen3:1.7b` and `qwen2.5-coder:7b` under identical case and scorer contracts. Thinking/reasoning modes were held consistent with preregistration. Primary aggregation: majority of three replicates per case.

4.3 AI and agent methodology disclosure

Frontier coding agents assisted with repository tooling and manuscript preparation; they are not listed as authors. All scientific endpoints come from preregistered local model executions with hashed prompts and responses. Routine editing does not constitute a reported experimental intervention.

4.4 Power and claim gates

E06 endpoints are known limited-power under the frozen case set. Confirmatory ordering claims require preregistered discordant pairs; exploratory secondary statistics are quarantined from primary promotion.

5 Results

Table 1 summarizes terminal verdicts for preregistered studies. Both experiments executed the full raw matrix ($n = 300$ cells each) with complete custody append.

EXP-008. Structured FCG retrieval did not yield promotable confirmatory evidence on E06 under the frozen design ($n = 2$ paired cases per model stratum at aggregation scope). Non-zero malformed and abstention rates are preserved in custody.

EXP-009. Primary verdict remains UNDERPOWERED for confirmatory E06 ordering. An exploratory secondary pattern was recorded as directionally positive at the descriptive layer only; it is **not promoted** to a causal claim.

Stage-2 baseline. Prior IC Failure Learning stage-2 execution (414 scored rows across two models) established `FAILURE_LEARNING_BEHAVIOR_IMPROVEMENT_NOT_ESTABLISHED`, providing historical context for the structured-retrieval falsification lane.

5.1 Broader experiment inventory (successor recovery)

This successor manuscript recovery catalogs **20** solo-scoped experiment lanes beyond the preregistered EXP-008/009 pair, including negative, blocked, and partial terminals (Appendix A). HydraLamp systems-validation results remain separated from treatment-effect claims (Table 2). Immersive Commons appears only as an operational hackathon portal, not as the scientific method.

5.2 Failure-preserving systems validation

The preregistered retrieval experiments above test whether structured context improves a frozen endpoint. A separate systems-validation suite asks a different question: does the custody mechanism

Table 2: Systems-validation outcomes (custody mechanics only; not treatment-effect evidence).

Validation	Scope	Outcome
Perturbation matrix	100 cells	100/100 chain verification
Synthetic tamper suite	8 modes	8/8 detected
Concurrent execution	10 runs	PASS (10 unique run IDs)
Replay/restart recovery	44 events	PASS
Live provider ladder	R0–R6	Bounded external failure preserved

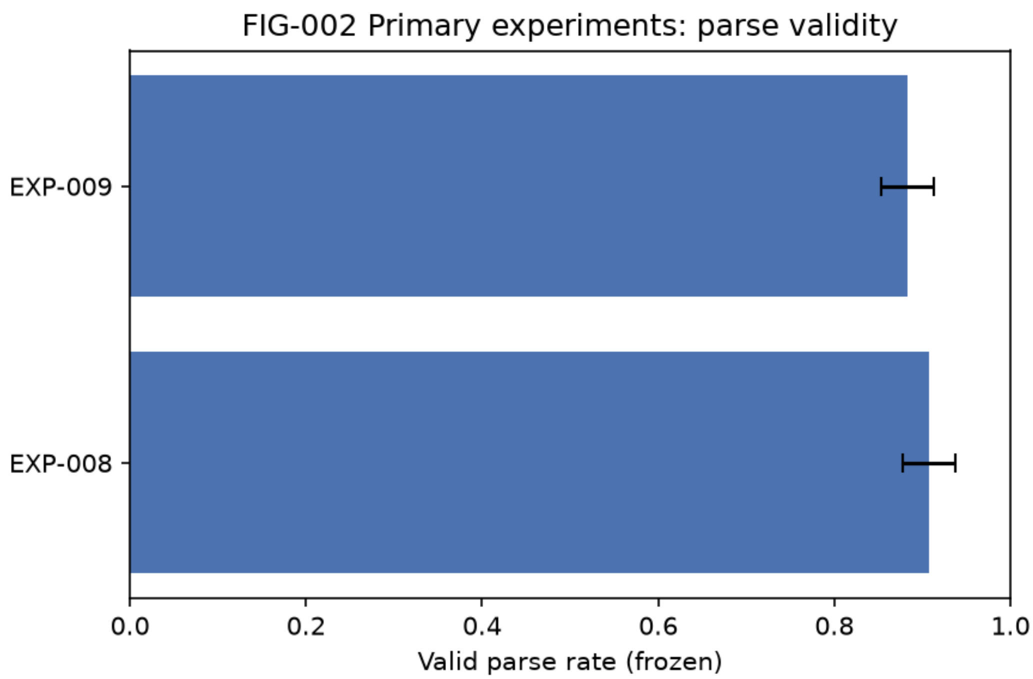

Figure 1: Primary experiment parse-validity rates (frozen observations). Confirmatory endpoints remain underpowered.

preserve integrity under perturbation, tampering, concurrency, replay, and external execution failure? Table 2 summarizes source-verified outcomes from a frozen HydraLamp perturbation matrix; these results evaluate custody mechanics and do **not** constitute evidence that structured FCG context improves the EXP-008/009 endpoint.

The tamper suite used synthetic cases only (not real security incidents). In a live provider repair ladder, earlier infrastructure gates passed before a structured call hit an external quota limit; the failure remained a terminal auditable state rather than being erased or substituted. The same custody primitives have been instantiated in sibling systems for local-model orchestration, deterministic scientific-document atomization, offline anomaly detection, bounded bioinformatics, substrate generation, and operator-facing scientific workflow monitoring; these implementations motivate transfer studies but do not provide additional evidence for the treatment effect reported here.

6 Discussion

HydraDG shows that agent experiments can retain null and failure cells with the negative-result integrity expected in preregistered experimental sciences: failures are first-class, hashes are mandatory, and claim ceilings block over-interpretation. The EXP-008/009 underpowered terminations bound what cannot yet be claimed while preserving malformed and timeout cells for audit.

6.1 Limitations

Limitations include bounded case replication, local-only model lanes in the reported matrix, and absence of runtime-equivalent successor replication in primary results. A preregistered Qwen 3.8 successor lane exists but is non-terminal and omitted from primary results. Whole-project hierarchical atomization (SeedGraph v1a validation) ran for multiple days but was interrupted during final Parquet serialization without a `BUILD_RECEIPT`; partial node/edge artifacts are not readback-safe, so terminal whole-corpus atomization is not established in this submission. Biological, protein-structure, and cellular perturbation applications of the custody framework remain future validation targets, not demonstrated generalizations of the agent-context results here.

6.2 Future directions

Agent systems. Larger successor replications, governed multi-agent handoffs, selective replay, and resource-aware custody gates under preregistered manifests.

Structured retrieval at scale. Checkpointed ingest and transactionally durable atomization so long hierarchy builds can resume from batch boundaries rather than materializing the entire graph at finalization.

Cross-project federation. Project-local FCG roots with explicit cross-project references and no automatic evidence admission between repositories.

Cryptographic custody. Authorized signatures and MMR commitments where policy permits; hashes remain identity commitments unless a verified signing operation occurred.

All directions above are *planned* unless independently measured; they do not extend the EXP-008/009 empirical claims.

7 Conclusion

We presented HydraDG, a custody-first framework for agent context-intervention experiments, and reported two terminal underpowered preregistered studies that block premature promotion of structured FCG retrieval effects on failure-prevention endpoints. We recommend FCO/FCG custody, explicit claim ceilings, and terminal-state preservation as baseline infrastructure for NewInML-style agent evaluations.

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A Full EXP-008 / EXP-009 statistical audit

Table 3 records terminal verdicts with parse rates reverse-traced to frozen verdict JSON receipts. p-values remain *not informative* under the preregistered underpowered design.

Table 3: Terminal preregistered study audit (appendix).

Study	Verdict	Raw cells	Valid parse rate
EXP-008	UNDERPOWERED	300	0.907
EXP-009	UNDERPOWERED	300	0.883

B Stage-2 M0/M1/M2 baseline

Stage-2 IC Failure Learning execution (414 scored rows across two models) closed as `FAILURE_LEARNING_BEHAVIOR_IMPROVEMENT_NOT_ESTABLISHED`; descriptive only.

C HydraLamp systems-validation matrix

See Table 4 for custody-mechanics outcomes (not treatment-effect evidence).

Table 4: Systems-validation matrix (appendix).

Validation	Scope	Outcome
Perturbation matrix	100 cells	100/100 chain verification
Synthetic tamper suite	8 modes	8/8 detected
Concurrent execution	10 runs	PASS (10 unique run IDs)
Replay/restart recovery	44 events	PASS
Live provider ladder	R0–R6	Bounded external failure preserved

D Deterministic verification and tooling

Figure 2 and `DETERMINISTIC_AUDIT_TOOLING_LEDGER.jsonl` document the custody verification pipeline with frozen script SHA-256 and R1/R2/R3 roots.

Source bytes → SHA-256 → Atom → ResultAtom → Claim/Table → FCG delta → exact-SHA verification

Figure 2: FIG-001: governed custody pipeline from source bytes to exact-SHA verification (structured data; self-generated).

E Citation and reference custody

Portfolio recovery bounded 185 occurrences to unique identities; only proposition-backed references are admitted. Citation fabrication literature informs desk-rejection risk [1], [2], [3], [4], [5].

F Requirement, template, and deadline drift

OpenReview license requirement CC BY 4.0 is frozen in requirement atoms; repository code remains Apache-2.0 (separate layer).

G Prior-art comparison matrix

Related-work boundaries include PROV-O, CWLProv, and Workflow Run RO-Crate [1], [2], [3], [4], [5]. Prior-art search responses remain *DISCOVERY_ONLY*; not promoted to verified empirical novelty proof.

H SeedGraph / FCO / FCG custody audit

`TOTAL_VERIFIED_INGEST_COMPLETE=NO` with verified coverage 31.55% (307/973).

I Planned and nonterminal lanes

GPU SGLang canary, Q38 XENV, and full verified ingest remain nonterminal.

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